

$$\frac{Q_{\mathrm{in}}}{L} \overset{??}{P}(t) \frac{t F_p F_a F_s F_d}{A_1 P_{\mathrm{out}}}$$

$$(1) \quad \frac{F_p(t) = [P(t) - P_{\mathrm{out}}] \cdot A_1}{A_1}$$

$$(2) \quad \frac{A_1 = \frac{\pi}{4} \left(r_1^2 - r_2^2 \right)}{r_1 \mathrm{m} r_2 \mathrm{m}} \underset{k}{}$$

$$(3) \quad \frac{G \cdot b \cdot h^3}{n \cdot D^3} \frac{\mathrm{GPa}}{b} \frac{b}{D} \frac{D}{n} \frac{\eta b/h}{F_s x(t)}$$

$$(4) \quad \frac{F_s(x) = k \cdot [x(t) + x_0]}{F_f}$$

$$(5) \quad \frac{F_d(t) = C \frac{dx(t)}{dt}}{C} \quad F_a$$

$$(6) \quad \frac{F_a(t) = m \frac{d^2 x(t)}{dt^2}}{m} \frac{Q_{\mathrm{in}} P(t) A_1 F_p}{x V(t) L A_o(x)} \underset{x <}{L x} \overset{>}{L A \cdot} \frac{\frac{dx}{dt}}{\frac{dx}{dt}}$$

$$(7) \quad \frac{Q = A_1 \frac{dx}{dt} + Q_n}{A_1 \frac{\mathrm{d} x}{\mathrm{d} t} Q_n \mathrm{m}^3/\mathrm{h}} + Q_n \frac{Q_n P(t) - P_{\mathrm{out}}}{P_{\mathrm{out}}}$$

$$(8) \quad Q_n = C_d A_n \sqrt{\frac{2 \left[P(t) - P_{\mathrm{out}} \right]}{\rho_w}} \frac{C_d A_n \mathrm{m}^2 \rho_w \mathrm{kg}/\mathrm{m}^3}{Q \gg} A_1 \frac{\mathrm{d} x}{\mathrm{d} t} Q_n$$

$$(9) \quad P(t) - P_{\mathrm{out}} = \Delta P_{\mathrm{close}}$$

$$(10) \quad (P(t) - P_{\mathrm{out}}) \cdot A_1 = \frac{A_1 \rho_2}{2 C_d^2 A_n^2} Q^2$$

$$(11) \quad \frac{d^2}{dt^2} + C_1 \left(\frac{dx}{dt} \right)^2 + (C_2 x + C_3) \frac{dx}{dt} + C_4 x + C_5 = 0$$

$$C_1$$

$$(12) \quad \frac{C_D A_2 \rho_w}{2 V \rho_f} C_2 C_3$$